Ashween Ramakrishnan

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LinkedIn | GitHub | Website | Google Scholar

SUMMARY

Software Engineer with more than four years of experience, including internships, building full-stack products across healthcare, medical devices, and enterprise learning platforms. Delivers features end-to-end, from database design to user interface, with a focus on performance, security, and reliable releases. Recent Master of Computer Science graduate (GPA 6.73/7.0) with hands-on experience applying AI, machine learning, and large language models to real-world problems.

SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP
- **Backend:** Laravel, Spring, Hibernate, Django, Apache Velocity, JSP
- **Frontend:** Vue.js, Nuxt.js, jQuery, HTML, CSS
- **AI / ML:** PyTorch, Hugging Face, LangChain, LangGraph, RAG, Scikit-learn, OpenCV
- **Databases & Cloud:** MySQL, MongoDB, AWS S3, REST APIs
- **DevOps & Developer Tools:** Docker, Jenkins, Git, SVN, Jira, Pytest, Ansible

EXPERIENCE

Allied AU - Software Engineer Intern Jul 2025 - Oct 2025
Tech Stack: Laravel (PHP), Nuxt.js, Vue.js, TypeScript, MySQL Adelaide, Australia

- Built **flexible pricing** on an allied-health platform, letting practitioners set separate Medicare, NDIS, and private rates for in-clinic, in-home, and telehealth sessions, end-to-end in **Laravel** and **Nuxt.js**.
- Developed **drag-and-drop appointment rescheduling** in the practitioner calendar, with a **Laravel API** that checks permissions and booking status, converts times to UTC, and auto-emails patients via **email templates**.
- Built **availability settings** for sole traders (new-client intake, waitlist, appointment limits, and session lengths) across **database**, **API**, and **onboarding**, so patients only book sessions within each practitioner's session lengths.
- Improved **NDIS** and **Medicare booking flows** by fixing booking and **validation errors**, adding NDIS provider checks, blocking in-home bookings when a patient lives outside the practitioner's travel radius, and redesigning the **NDIS earnings view** to show practitioners their net earnings.
- Improved the user experience for patients, practitioners, and businesses with a post-approval welcome page, clearer onboarding navigation, **responsive mobile layouts** for sign-up and sign-in, a working guided tour for new businesses, and **booking search** that matches multi-word queries such as full names.
- Shipped **49 changes** across **Laravel** and **Nuxt.js** via **GitLab CI/CD** and code review, working with the CEO, product manager (QA), and senior developer (DevOps) on bug fixes, staging validation, and releases.

LinkedIn - Software Engineer/Contingent Worker Jul 2022 - Jun 2024
Tech Stack: Java, JSP, JavaScript, Hibernate, SQL India

- Built **checkout**, **subscription**, and **invoice workflows**, and the **Trailhead integration** for Salesforce's Trailhead Academy LMS using **Java**, **JavaScript**, and **CyberSource**, including **pricing schemes** connected to an external pricing service, **coupon** and **currency handling**, automated purchase emails, and **refund-aware invoicing**.
- Implemented **subscription validation**, **access control**, and **course filtering**: eligibility checks before event sign-up, hiding subscription-only events from inactive subscribers, configurable catalogue settings, and subscription, content-group, empty-value, and invoice filters that improve **course discoverability**.
- Fixed slow **Guided Path** and **Order History** pages by analysing **query logs**, rewriting inefficient **database queries**, and cutting **repeated database calls**.
- Closed a **DOM-based Cross-Site Scripting (XSS)** vulnerability, where an unvalidated callback URL parameter in a shared **JSP** page let attackers inject script.
- Built **user** and **site data deletion** across domain objects and automated export and **migration scripts** (DocuSign, course files, bulk admin onboarding), supporting privacy compliance and secure customer offboarding.

- Resolved **enterprise customer tickets** with bug fixes across checkout, payments, subscriptions, and course listings, and took part in **code reviews**.

EduBrite Systems, Inc. (now part of LinkedIn) - Full Stack Developer Oct 2020 - Jun 2022
Tech Stack: Java, Spring, Hibernate, Apache Velocity, JSP, JavaScript India

- Developed custom LMS modules for enterprise clients **Salesforce** and **VMware** using **Java**, **Spring**, **Hibernate**, **Apache Velocity**, **JavaScript**, **HTML**, and **CSS**, and packaged shared libraries into **platform releases**.
- Designed an **event-driven email notification engine** in **Java** and **Apache Velocity** that automatically sends learners and instructors event-specific emails, with admin controls, and fixed **production email issues**.
- Extended the **Velocity templating engine** by exposing backend helpers and **utility APIs** (custom configuration **JSON**, time zones, localised text) to templates, so each customer's emails and pages could be customised by editing templates alone, **without new code deployments**.
- Built learning and event-catalogue features: **learner attendance tracking** with grading, completion progress, group-based **access filtering**, and search, sort, and filtering by custom attributes, status, time zone, and location.
- Built **QR codes** on assessment pages using **JavaScript** to help identify users who **leak** or **publish** test questions, with an option in the test editor so admins could switch it on for individual tests.
- Improved **performance** and commerce flows by cutting **repeated database calls** when loading course and program resources, and enhanced **cart** and **invoice pages** with advanced filters and DocuSign-linked invoices.

ICU Medical India LLP - Intern, Embedded Team (Python Test Automation) Feb 2020 - Jun 2020
Tech Stack: Python, Pytest, Linux, Jenkins India

- Developed **Python-based** automated test suites using **Pytest** to validate **12 therapy features** of the Plum 360 Smart Infusion Pump, improving the reliability of **safety-critical** medical software used in clinical environments.
- Designed **nominal** and **edge-case test** scenarios validating UI widget interactions, **communication APIs**, safety pop-ups, alarms, and drug-limit alerts, strengthening product safety and regulatory compliance.
- Increased line coverage from **54% to 72%** (measured on **Jenkins**) with reusable Pytest modules running on a **Linux-based** pump simulator via layered **Python APIs**, tracked in **Jira** in a regulated medical-device workflow.

PROJECTS

- **Smart Contract Vulnerability Detector:** Fine-tuned an **LLM classifier** that detects reentrancy vulnerabilities with **91.6%** accuracy, built a control-flow visualiser, and wrote a YAML-based **Ansible playbook** that automates **AWS** cloud deployment of the end-to-end tool into Adelaide University's CyberLab platform.
- **Cybersecurity Intelligence System:** Built a Retrieval-Augmented Generation (**RAG**) pipeline that stores scraped MITRE ATT&CK data in a **ChromaDB** vector store and feeds relevant entries to a compressed **Mistral-7B LLM**, generating grounded threat mitigations with fewer hallucinations, evaluated with **RAGAS** metrics.
- **LiveWell - Elderly Care Companion:** Built a **React Native app** with geofencing alerts, accelerometer fall detection, and a voice assistant (**LangGraph**, Flask, Whisper, Llama) for daily tasks and caregiver alerts.

EDUCATION

The University of Adelaide (now part of Adelaide University) **Adelaide, Australia**
Master of Computer Science | GPA: 6.73/7.0 Jul 2024 - Oct 2026
Amrita Vishwa Vidyapeetham **Coimbatore, India**
Bachelor of Technology in Computer Science and Engineering Jul 2016 - Jun 2020

CERTIFICATIONS, CONTRIBUTIONS & LEADERSHIP

- **Certifications:** Microsoft Certified: Azure Fundamentals, Docker Foundations Professional Certificate
- **Open-Source:** OpenSeadragon - Contributed to the open-source image viewer by enhancing zoom-control APIs and DOM event handling, with multiple PRs merged into the core project.
- **Leadership:** Director of Events and Officer of Sports and Wellbeing, Master's in Technology Student Association

— References available on request —